

# Chapter 9 - The Voodoo 3D Rasterizer

Voodoo is Intuition Engine's hardware triangle rasterizer. It draws flat-shaded, Gouraud-shaded, and textured triangles into an RGBA framebuffer with an associated Z-buffer. Voodoo sits on compositor layer 20, the top of the stack, so its picture covers every other source where it draws.

This chapter describes Voodoo's programming model. The chip's register set is closely modelled on the 3DFX Voodoo Graphics SST-1, so existing 3DFX programming literature is a useful reference - but the addresses are IE addresses, given in the tables below.

## 9.1 What Voodoo can show

| Item               | Value                                                                   |
|--------------------|-------------------------------------------------------------------------|
| Output             | RGBA framebuffer with depth (Z) buffer                                  |
| Default resolution | 640 × 480                                                               |
| Maximum resolution | 800 × 600                                                               |
| Primitive          | Triangle, defined by three vertices                                     |
| Per-vertex data    | Position (12.4), RGBA (12.12), Z (20.12), S/T texture (14.18), W (2.30) |
| Texture sizes      | up to 256 × 256 from 64 KB of texture memory                            |
| Texture formats    | 11 formats including paletted, ARGB1555, ARGB4444, ARGB8888             |
| Depth functions    | 8 (NEVER through ALWAYS)                                                |
| Alpha test/blend   | Yes, 16 blend factors                                                   |
| Fog                | Per-pixel, table-based                                                  |
| Chroma key         | Yes                                                                     |

## 9.2 Programming model

Voodoo is **state-machine driven**. A program prepares the chip by writing into mode and state registers, builds a triangle by writing three vertex positions and a set of "start" colour and texture values, and finally issues a triangle command. The chip then rasterises the triangle into the back buffer.

The shortest path from machine language:

1. Enable the chip (VOODOO\_ENABLE = 1).
2. Pick the output dimensions in VOOODO\_VIDEO\_DIM.
3. Configure FBZ\_MODE, ALPHA\_MODE, and FBZCOLOR\_PATH for the desired pixel-pipeline behaviour.
4. (Optional) Upload textures into texture memory and configure TEXTURE\_MODE and the texture base registers.
5. For each triangle: - Write vertex positions to VERTEX\_AX/AY/BX/BY/CX/CY. - Write the start values (colour, Z, S, T, W) and any per-X and per-Y deltas. - Trigger rasterisation by writing to TRIANGLE\_CMD.
6. Swap the buffers with SWAP\_BUFFER\_CMD.

## 9.3 BASIC keywords

The V00D00 keyword introduces every Voodoo subcommand. A companion VERTEX, TRIANGLE, ZBUFFER, and TEXTURE are recognised as standalone hardware verbs that operate on the same state. See Chapter 2 for complete entries.

| Form                                                                                                                                                                 | Effect                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| V00D00 ON / V00D00 OFF                                                                                                                                               | Enable / disable the chip.                                                                                                                                   |
| V00D00 DIM <sub>w,h</sub>                                                                                                                                            | Set the output dimensions.                                                                                                                                   |
| V00D00 CLEAR <sub>r,g,b</sub>                                                                                                                                        | Fast-fill the back buffer with a colour.                                                                                                                     |
| V00D00 SWAP                                                                                                                                                          | Swap front and back buffers.                                                                                                                                 |
| V00D00 CLIP <sub>x0,y0,x1,y1</sub>                                                                                                                                   | Set scissor rectangle.                                                                                                                                       |
| V00D00 COMBINE <sub>mode</sub>                                                                                                                                       | Set the colour-combine mode (ITERATED/TEXTURE/MODULATE/ADD/DECAL).                                                                                           |
| V00D00 LFB <sub>addr</sub>                                                                                                                                           | Set the linear-framebuffer base for LFB writes.                                                                                                              |
| V00D00 PIXEL <sub>x,y,r,g,b</sub>                                                                                                                                    | Write one pixel through the LFB port.                                                                                                                        |
| V00D00 RGB ON / OFF                                                                                                                                                  | Enable / disable RGB writes.                                                                                                                                 |
| V00D00 TRICOLOR <sub>r,g,b[,a]</sub>                                                                                                                                 | Set the per-triangle start colour at vertex A (writes START_R, START_G, START_B, and optionally START_A). For flat shading, leave the colour deltas at zero. |
| V00D00 TRISHADE <sub>drdx,drdy,dgdx,dgdy,dbdx,dbdy</sub>   Set the six per-pixel colour gradients for Gouraud shading (writes DRDX/DRDY/DGDX/DGDY/DBDX/DBDY).        |                                                                                                                                                              |
| V00D00 TRIDEPTH <sub>start_z,dzdx,dzdy</sub>   Set the depth start value and gradients (writes START_Z,DZDX,DZDY).                                                   |                                                                                                                                                              |
| V00D00 TRIUV <sub>start_s,start_t,dsdx,dtdx,dsdy,dt dy</sub>   Set the texture coordinate start and gradient registers (writes START_S,START_T,DSDX,DTDY,DSDY,DTDY). |                                                                                                                                                              |
| V00D00 TRIW <sub>start_w,dwdx,dwdy</sub>                                                                                                                             | Set the perspective W (1/Z) start value and gradients (writes START_W,DWDY,DWDY).                                                                            |
| V00D00 ALPHA <sub>mode</sub>                                                                                                                                         | Configure the alpha pipeline (test enable/function, blend enable/factors).                                                                                   |
| V00D00 FOG ON / OFF / COLOR <sub>r,g,b`</sub>                                                                                                                        | Configure fog.                                                                                                                                               |
| V00D00 DITHER ON / OFF                                                                                                                                               | Enable / disable dithering.                                                                                                                                  |
| V00D00 CHROMAKEY ON / OFF / COLOR <sub>r,g,b`</sub>                                                                                                                  | Configure chroma key.                                                                                                                                        |
| VERTEX <sub>A<sub>x,y</sub>/VERTEX B ... / VERTEX C ...`</sub>                                                                                                       | Write a vertex's screen position.                                                                                                                            |
| TRIANGLE                                                                                                                                                             | Submit the current vertex set as one triangle.                                                                                                               |
| ZBUFFER ON / OFF / FUNC <sub>n</sub> /WRITE <sub>flag</sub>                                                                                                          | Configure depth testing and depth writes.                                                                                                                    |
| TEXTURE <sub>args</sub>                                                                                                                                              | Upload texture data (Chapter 2 lists all forms).                                                                                                             |

# 9.4 The register block

Voodoo's register block runs 0xF8000–0xF87FF. Registers are 32-bit words at 4-byte-aligned addresses.

## 9.4.1 Control and status

| Address | Name          | Purpose             |
|---------|---------------|---------------------|
| 0xF8000 | V00D00_STATUS | Status (read-only). |
| 0xF8004 | V00D00_ENABLE | Bit 0 = enable.     |

V00D00\_STATUS bits:

| Bits  | Field    | Meaning                       |
|-------|----------|-------------------------------|
| 0     | FBI_BUSY | Framebuffer interface busy.   |
| 1     | TMU_BUSY | Texture-mapping unit busy.    |
| 2     | SST_BUSY | Overall chip busy.            |
| 6     | VRETRACE | Vertical retrace active.      |
| 7     | SWAPBUF  | Buffer-swap pending.          |
| 12–19 | MEMFIFO  | Memory-FIFO entries (8 bits). |
| 20–24 | PCIFIFO  | PCI-FIFO entries (5 bits).    |

## 9.4.2 Vertex registers

The six vertex coordinate registers hold three pairs of (X, Y) in 12.4 fixed-point format. The "start" registers hold the values of colour, depth, and texture coordinates at vertex A; the eight "dX" registers hold the per-X-step derivatives, and the eight "dY" registers hold the per-Y-step derivatives.

| Address         | Name          | Format                |
|-----------------|---------------|-----------------------|
| 0xF8008–0xF801C | VERTEX_AX–CY  | 12.4 fixed-point      |
| 0xF8020         | START_R       | 12.12 fixed           |
| 0xF8024         | START_G       | 12.12 fixed           |
| 0xF8028         | START_B       | 12.12 fixed           |
| 0xF802C         | START_Z       | 20.12 fixed           |
| 0xF8030         | START_A       | 12.12 fixed           |
| 0xF8034         | START_S       | 14.18 fixed           |
| 0xF8038         | START_T       | 14.18 fixed           |
| 0xF803C         | START_W       | 2.30 fixed            |
| 0xF8040–0xF805C | DRDX ... DWDX | 8 × per-X derivatives |
| 0xF8060–0xF807C | DRDY ... DWDY | 8 × per-Y derivatives |

### 9.4.3 Command registers

| Address | Name            | Purpose                                                                           |
|---------|-----------------|-----------------------------------------------------------------------------------|
| 0xF8080 | TRIANGLE_CMD    | Rasterise one triangle using the current state.                                   |
| 0xF8084 | FTRIANGLECMD    | Fast triangle (triangle-strip mode).                                              |
| 0xF8088 | COLOR_SELECT    | Select the vertex (0/1/2) that subsequent colour writes target (Gouraud shading). |
| 0xF8120 | NOP_CMD         | No operation.                                                                     |
| 0xF8124 | FAST_FILL_CMD   | Fast-fill the buffer with COLOR0.                                                 |
| 0xF8128 | SWAP_BUFFER_CMD | Swap front/back buffers (bit 0 = wait for VSync, bit 1 = clear after swap).       |

### 9.4.4 Mode and state

| Address | Name            | Purpose                                              |
|---------|-----------------|------------------------------------------------------|
| 0xF8104 | FBZCOLOR_PATH   | Colour combine and source selects.                   |
| 0xF8108 | FOG_MODE        | Fog pipeline configuration.                          |
| 0xF810C | ALPHA_MODE      | Alpha test/blend pipeline.                           |
| 0xF8110 | FBZ_MODE        | Framebuffer Z mode (depth + clipping + draw enable). |
| 0xF8114 | LFB_MODE        | Linear-framebuffer access mode.                      |
| 0xF8118 | CLIP_LEFT_RIGHT | Scissor left/right (10-bit each).                    |
| 0xF811C | CLIP_LOW_Y_HIGH | Scissor top/bottom (10-bit each).                    |

FBZ\_MODE (low 21 bits):

| Bit | Field        | Meaning                           |
|-----|--------------|-----------------------------------|
| 0   | CLIPPING     | Enable scissor.                   |
| 1   | CHROMAKEY    | Enable chroma keying.             |
| 2   | STIPPLE      | Enable stipple.                   |
| 3   | WBUFFER      | Use W buffer instead of Z.        |
| 4   | DEPTH_ENABLE | Enable depth test.                |
| 5–7 | DEPTH_FUNC   | Depth compare function (0–7).     |
| 8   | DITHER       | Enable dithering.                 |
| 9   | RGB_WRITE    | Enable RGB writes.                |
| 10  | DEPTH_WRITE  | Enable depth writes.              |
| 11  | DITHER_2X2   | 2×2 dither (vs 4×4).              |
| 12  | ALPHA_WRITE  | Enable alpha writes.              |
| 14  | DRAW_FRONT   | Draw to front buffer.             |
| 15  | DRAW_BACK    | Draw to back buffer.              |
| 16  | DEPTH_SOURCE | 0 = iterated, 1 = floating-point. |

| Bit   | Field        | Meaning                        |
|-------|--------------|--------------------------------|
| 17    | Y_ORIGIN     | 0 = top-left, 1 = bottom-left. |
| 18    | ALPHA_PLANES | Enable alpha planes.           |
| 19    | ALPHA_DITHER | Dither alpha.                  |
| 20–31 | DEPTH_OFFSET | 12-bit depth bias.             |

The depth and alpha functions share the same encoding:

| Value | Function     |
|-------|--------------|
| 0     | NEVER        |
| 1     | LESS         |
| 2     | EQUAL        |
| 3     | LESSEQUAL    |
| 4     | GREATER      |
| 5     | NOTEQUAL     |
| 6     | GREATEREQUAL |
| 7     | ALWAYS       |

ALPHA\_MODE low bits:

| Bit   | Field     | Meaning                         |
|-------|-----------|---------------------------------|
| 0     | TEST_EN   | Enable alpha test.              |
| 1–3   | TEST_FUNC | Alpha test function.            |
| 4     | BLEND_EN  | Enable alpha blending.          |
| 5     | ANTIALIAS | Enable antialiasing.            |
| 8–11  | SRC_RGB   | Source RGB blend factor.        |
| 12–15 | DST_RGB   | Dest RGB blend factor.          |
| 16–19 | SRC_A     | Source alpha blend factor.      |
| 20–23 | DST_A     | Dest alpha blend factor.        |
| 24–31 | REF       | Reference value for alpha test. |

The 16 blend-factor codes follow the 3DFX convention (0 = zero, 1 = source alpha, 4 = one, 5 =  $1 - \text{src.A}$ , etc.).

## 9.4.5 Fog, chroma, stipple, constants

| Address         | Name                                         | Purpose                     |
|-----------------|----------------------------------------------|-----------------------------|
| 0xF8140–0xF823C | FOG_TABLE_BASE (64 entries $\times$ 4 bytes) | Per-zone fog blend factors. |
| 0xF81C4         | FOG_COLOR                                    | Fog RGB.                    |
| 0xF81C8         | ZA_COLOR                                     | Z/A constant.               |
| 0xF81CC         | CHROMA_KEY                                   | Chroma key colour.          |

| Address | Name         | Purpose                                    |
|---------|--------------|--------------------------------------------|
| 0xF81D0 | CHROMA_RANGE | Chroma key tolerance.                      |
| 0xF81D4 | STIPPLE      | Stipple pattern.                           |
| 0xF81D8 | COLOR0       | Constant colour 0 (used by FAST_FILL_CMD). |
| 0xF81DC | COLOR1       | Constant colour 1.                         |

## 9.4.6 Display timing and configuration

| Address         | Name            | Purpose                     |
|-----------------|-----------------|-----------------------------|
| 0xF8200–0xF8210 | FBI_INIT0–INIT4 | Framebuffer-interface init. |
| 0xF8214         | VIDEO_DIM       | width << 16 \   height.     |
| 0xF8218         | BACK_PORCH      | Back-porch timing.          |
| 0xF821C         | VIDEO_DIM_V     | Vertical dimensions.        |
| 0xF8220         | H_SYNC          | Horizontal sync.            |
| 0xF8224         | V_SYNC          | Vertical sync.              |
| 0xF822C         | DAC_DATA        | DAC data port.              |

## 9.4.7 Texture unit

| Address         | Name                       | Purpose                                          |
|-----------------|----------------------------|--------------------------------------------------|
| 0xF8300         | TEXTURE_MODE               | Texture pipeline configuration.                  |
| 0xF8304         | TLOD                       | LOD control.                                     |
| 0xF8308         | TDETAIL                    | Detail texture control.                          |
| 0xF830C–0xF832C | TEX_BASE0–TEX_BASE8        | Texture base addresses per LOD.                  |
| 0xF8330         | TEX_WIDTH                  | Texture upload width (IE extension).             |
| 0xF8334         | TEX_HEIGHT                 | Texture upload height (IE extension).            |
| 0xF8338         | TEX_UPLOAD                 | Write to commit a texture upload (IE extension). |
| 0xF8400–0xF87FF | PALETTE_BASE (256 entries) | Texture palette.                                 |

TEXTURE\_MODE low bits:

| Bit  | Field   | Meaning                       |
|------|---------|-------------------------------|
| 0    | ENABLE  | Enable texturing.             |
| 1–3  | MINIFY  | Minification filter.          |
| 4    | MAGNIFY | 0 = point, 1 = bilinear.      |
| 5    | CLAMP_S | Clamp S.                      |
| 6    | CLAMP_T | Clamp T.                      |
| 8–11 | FORMAT  | Texture format (table below). |
| 12   | CHROMA  | Texture chroma key.           |

| Bit | Field       | Meaning                 |
|-----|-------------|-------------------------|
| 13  | TRILINEAR   | Trilinear filtering.    |
| 14  | PERSPECTIVE | Perspective correction. |
| 15  | DETAIL      | Detail texturing.       |
| 16  | SEQUENCE    | Sequence enable.        |

Texture formats:

| Code | Format         |
|------|----------------|
| 0    | 8-bit paletted |
| 1    | YIQ compressed |
| 2    | A8             |
| 3    | I8             |
| 4    | AI44           |
| 5    | P8             |
| 6    | ARGB8332       |
| 7    | AI88           |
| 8    | ARGB1555       |
| 9    | ARGB4444       |
| 10   | ARGB8888       |

## 9.5 Texture memory

Voodoo's texture memory is a separate 64 KB region beginning at 0xD0000. Textures are uploaded by writing pixel data into this region, then setting TEX\_WIDTH/TEX\_HEIGHT and writing to TEX\_UPLOAD to commit. The chip then samples the texture during triangle rasterisation according to the TEXTURE\_MODE and the S/T/W start values and derivatives.

## 9.6 The colour pipeline

FBZCOLOR\_PATH selects how the iterated (vertex) colour, the texture colour, and the constant colour are combined into the final pixel. The two-source select and the combine mode are packed into one register:

| Bits | Field          | Meaning                                         |
|------|----------------|-------------------------------------------------|
| 0–1  | RGB source     | 0 = iterated, 1 = texture, 2 = COLOR1, 3 = LFB. |
| 2–3  | Alpha source   | Same encoding as RGB.                           |
| 4–6  | Combine mode   | One of eight functions (see table below).       |
| 27   | TEXTURE_ENABLE | Include texture colour in the path.             |

| Combine code | Function | Meaning       |
|--------------|----------|---------------|
| 0            | ZERO     | Output black. |

| Combine code | Function | Meaning                   |
|--------------|----------|---------------------------|
| 1            | CSUB_CL  | other - local.            |
| 2            | ALOCAL   | local × local_alpha.      |
| 3            | AOTHER   | local × other_alpha.      |
| 4            | CLOCAL   | local (pass through).     |
| 5            | ALOCAL_T | local_alpha × texture.    |
| 6            | CLOC_MUL | local × other (modulate). |
| 7            | AOTHER_T | other_alpha × texture.    |

## 9.7 Putting it together

The minimum machine-language sequence to draw one solid-colour triangle from (100, 100) to (200, 100) to (150, 200) at full red:

1. Write 1 to V00D00\_ENABLE.
2. Write (640 << 16) | 480 to VIDEO\_DIM.
3. Write a value into FBZ\_MODE that enables RGB writes, depth writes, and draw to back buffer.
4. Write 0 to FBZCOLOR\_PATH (iterated colour, no texture).
5. Write 100 << 4 to VERTEX\_AX, 100 << 4 to VERTEX\_AY, 200 << 4 to VERTEX\_BX, 100 << 4 to VERTEX\_BY, 150 << 4 to VERTEX\_CX, 200 << 4 to VERTEX\_CY.
6. Write 255 << 12 to START\_R; zero to START\_G, START\_B, and the eight delta registers.
7. Write any value to TRIANGLE\_CMD.
8. Write 1 to SWAP\_BUFFER\_CMD to swap with the next VSync.

The next chapter shows how the same chip and its companions can build tile and sprite layers on top of the other picture sources.